18. Environmental Air Pollution

Beyond the Plant

Often the tendency is to assume that once the contaminant has been removed from the plant, the responsibilities of the industrial hygiene engineer are complete. In the past, the practice has been to discharge contaminants in the air outside the plant and depend upon natural dilution to remove the hazard. However, this situation can create problems. Natural dilution does not always exist. During periods of weather inversions, contaminants can build up to a point where a hazard or nuisance to the surrounding area is created. The industrial hygiene engineer should consider the effects of pollution to the surrounding area and should determine the extent of pollution that occurs. Controls should be developed to minimize the potential hazard that can result from environmental air pollution.

Air pollution is not limited to that which is produced by industrial processes. There are certain natural types of air pollution which have existed since the beginning of time. Dust that is picked up and moved from area to area is an example. The dust bowl that occurred during the mid-thirties was an extreme case of such natural air pollution. Other sources of natural air pollution include smoke resulting from naturally caused fires and pollen that floats through the air, raising havoc among certain people. Chemical reactions that occur in nature also produce contaminants which pollute the atmosphere. However, for the most part, these natural sources of air pollution are beyond the control of man. Also, since they have existed from the beginning of time, certain defense mechanisms have been built up within man to protect against this type of pollution.

On the other hand, man-made pollution is a relatively new phenomenon. The great majority of pollution that exists in the atmosphere today has probably come into being only since the beginning of the industrial revolution. Man-made pollutants include the exhaust from automobiles and emissions from fuel and energy production. Heat production at local plants as well as homes also pollutes the atmosphere. Industrial processes and open burning can be a major source of hazardous pollutants.

The Effects of Air Pollution. The presence of air pollution can result in many harmful effects. Among the most important of these effects are:

1. Health effects--This area has become the subject of much controversy. Statements have been made that cancer is related to industrial pollutants. The accuracy of such statements may be subject to question; however, there are documented cases where acute

air pollution has resulted in excessive deaths. Certain chronic effects have been studied. These studies seem to indicate that a cause/effect relationship between air pollution and health does exist. Further studies are being conducted, and it is likely that this subject will remain a controversial one for the foreseeable future.

2. Community relations--Excessive air pollution can be harmful to community relations for the industries involved. The workers may react to the dangers that threaten their families' health if they live in the immediate area. Community groups may form to put pressures on the industry to reduce the amount of pollution present. In addition, the public relations image obtained by the plant involved may be harmed, resulting in a negative effect on sales and profits.

3. Property damage--One needs only to drive around a heavily industrialized area to see indications of the property damage that can occur as a result of air pollution. The economic effects of this damage are hard to determine. Equipment may operate over a shorter life because of corrosion caused by pollution. Buildings must be cleaned more often, and the exterior surfaces may be subject to excessive aging.

4. Economic climate of the community--Excessive air pollution can affect the economic climate of the community. New businesses are less likely to move into a dirty area because of the difficulty in finding workers willing to live in the vicinity of the plant.

5. Plant life--Botanists have done studies on the effects of air pollution on various types of plant life. Direct cause and effect relationships have been discovered for certain types of pollution.

6. Animal life--As in the case of health factors affecting humans, certain acute pollution problems have been shown to exist in animals. Currently, the controversy surrounding PCB's and Kepone indicate the possibility of long-range chronic effects of pollution on animals, not to mention the potential second-level effects on man. As a part of the natural food chain, pollutants that affect plant life are often carried over to animals and thus to humans. The chronic effects of this type of transfer are not well known or understood.

Determining the Extent of Air Pollution--Measuring the Ambient Air

There are two methods that are available to determine the extent of air pollution. The first of these methods is the measuring of the ambient air. Such measurement involves determining the extent and dispersion of pollution in the surrounding area. Pollution is affected by the weather patterns and topography present. In addition, any synergistic effects between two pollutants from different industrial plants can also be considered by this method. Measuring the ambient air is somewhat akin to measuring the

contaminant within the workplace. Though it does not give specific information related to the pollutants from a particular source, it does present information concerning the potential hazards that may be present in the atmosphere.

The sampling train that is used in measuring the ambient air includes the following items:

1. Collector for contaminants.
2. Air mover or vacuum source.
3. Airflow measuring device.
4. Sample probe or collection device.
5. Equipment to measure other factors in the environment, such as heat, humidity, and pressure.

The type of collection device that is used in the sampling train depends on the type of pollutant that is present. The basic types of pollutants are gases and vapors or particulate matter. Collection devices for gases and vapors are generally of two types. Either a grab sample (without concentration of contaminant) is taken, or a continuous sample with concentration of contaminant is taken. For grab samples, an evacuated container or a liquid displacement method is used. Continuous collection with concentration involves either absorption in liquids (both chemical and physical absorption), adsorption on solids such as activated charcoal, or condensation or freeze-out methods.

Collection devices in the sampling train for particulate matter involve sedimentation, inertial separation, electrostatic precipitation, thermal precipitation, filtration, or the use of photometers. It is beyond the scope of this discussion to go into the various advantages and disadvantages of each of these methods. The reader is referred to The Air Pollution Manual, Part I, Evaluation, published by the American Industrial Hygiene Association for further study and references on this subject. In general, the methods are similar to the methods for cleaning air previously discussed.

Certain observations of the smoke plume can also give some estimate of the amount of air pollution that is present. The Ringelmann chart has been used in the past for such observations. In addition, observers can be trained to estimate the pollution due to particulate matter by observing the smoke plume.

The second major component of a sample train is the air mover. The purpose of the air mover is to create a flow of air through the collector. Two major types of air movers are used--the pump and the ejector.

There are various types of pumps that are used in such sample trains. The piston-and-diaphragm pump is commonly used. In addition, pumps may be powered by a motor or by a hand crank. When small grab samples are desired, the system is often powered by a squeeze bulb. Ejectors that work on the venturi principle are often used in an area where the atmosphere is explosive.

Some method must be present in the sampling train to meter the amount of air that is being drawn into the collector. To determine the concentration of contaminant, it is necessary that the amount of sample air is known. A number

of measuring devices are available, including the dry gas meter, the wet test meter, and the cycloid type meter. These methods measure the quantity of air being delivered. The rate of flow can be measured by the use of head meters, including the venturi, flow nozzle, and orifice types. Rate of flow can also be measured by laminar flow meters and rotameters.

In any case, each of the above methods requires a calibration to assure that the proper metering occurs. The spirometer is a primary standard against which all meters are calibrated. Secondary standards of calibration are often used. This method involves calibration of a meter against a primary standard. This meter, which is generally more accurate, is then used to calibrate a second type of meter.

Other equipment necessary for the sampling train includes the sampling probe which must be inert to contaminants and resist clogging by particulate matter. It is also necessary that an accurate method for measuring the temperature, humidity, pressure, and time be provided. Valves to regulate flow rate should also be present in the sampling train.

Among the factors that should be considered when taking a sample are the following:

1. How long to sample.
2. How much to sample.
3. Where to sample.
4. The methods to be used to obtain the sample.
5. The methods of sample analysis.
6. The storage of the sample.
7. The handling of the sample.
8. The influence of variables such as weather, topography, and other factors on the representativeness of the sample.

Determining the Extent of Pollution--Measuring at the Source

A second method to determine the extent of pollution is to measure the pollution that is resulting from a single source. This provides information concerning the pollutants that are leaving the plant. From this information, a determination can be made as to whether the methods of control used within the plant are adequate.

When it is decided that measurements at the source should take place, the industrial hygiene engineer must consider all sources of pollutants that may escape from the plant. These pollutants may result from industrial processes or may be the result of service equipment such as boiler equipment. All outlets to the atmosphere and vents from exhaust system or stacks must be monitored to determine the total extent of the pollution that is entering the atmosphere from the source.

The procedure for obtaining measurements of pollutants at the source involves, first, an identification of the pollutant. Is the pollutant a gas, vapor, or particulate matter? What is the chemical composition of the pollutant? Do any chemical reactions occur during exhaust that must be

considered? It may be necessary that a qualitative analysis of the emission be taken to determine the type of pollutants that are being exhausted.

The industrial hygiene engineer should determine where the sample is to be taken. It can be taken in the exhaust system prior to the fan, or it may be taken in the stack or at the vents. The location chosen depends on the particular situation involved. In some cases, it may be difficult to take measurements within the stack, while in other cases these measurements may yield a combined pollutant resulting from a chemical reaction that is unknown if measurements are taken in each of the individual exhaust ducts.

The next step is to determine the velocity of the air traveling through the duct or in the stack. In order to ascertain this, a traverse of the duct or stack is necessary. A pitot tube or anemometer can be used for this traverse, depending upon the velocity and temperature of the air within the duct. Using the average velocity obtained and a measurement of the diameter of the outlet, the industrial hygiene engineer can determine the rate of flow at which the pollutant is entering the surrounding atmosphere.

A determination should be made of the pressure, gas, temperature, and moisture content that is present in the emission. These factors will affect the rate of flow if significantly different from nonstandard conditions.

As in the case of sampling ambient air, it is necessary to develop a sampling train to collect and measure the sample from the source. If particulate matter is involved, it is desirable to determine the expected particle size since, unless special care is taken, agglomeration as well as particle breakup can occur during the collection process. Filtration is commonly used to collect particulate contaminants within the stack. A cyclone dust separator may also be used prior to the filter to remove large particles. Other methods that may be used include impingement and electrostatic precipitation. If a gas or vapor is present, the collection can be through either adsorption or absorption equipment as in measuring ambient air. Air movers and metering devices are also required when sampling from the source. Movers are much like the pumps and ejectors that are used in sampling ambient air, and the air metering methods are also the same. However, it is necessary that the velocity of the sampling train match that of the duct or stack (isokinetic conditions). This is often accomplished by using a static balanced tube or probe or by adjusting the sampling rate and nozzle size to match the velocity in the duct or stack. If these precautions are not taken and the duct or stack velocity differs significantly from that of the sampling train, then either too much or too little air will be sampled.

After the collection of the sample, a measurement and analysis must be made to determine the quantity and chemical composition of the materials gathered. These results can be compared to regulated standards and/or desired results to determine if the proper control is being exercised.

Other observations can be made concerning the plume appearance, the presence of water droplets in gas, the presence of deposits in the stack, and the nature of these droplets. In addition, the production schedule for the process producing the contaminants should be considered. Some production facilities operate on a cyclic basis while others are continuous. Obviously,

the timing of the sample must coincide with the production schedule. Sampling during the worst possible conditions will provide a measure of safety in terms of errors made and pollutants emitted to the atmosphere.

Control of Environmental Air Pollution

This subject has been discussed in much detail in the previous chapter concerning air-cleaning devices. However, it is useful to review the general methods that are available. The basic types of air cleaners are--

Cleaners for Particulate Matter

- Mechanical collectors
  - Settling chambers
  - Cyclones
  - Impingers
  - Dynamic precipitators
- Filtration
  - Deep-bed filters
  - Fabric filters (bag houses)
  - High-efficiency panel filters
- Wet collectors
  - Cyclone scrubbers
  - Chamber scrubbers
  - Self-induced spray scrubbers
  - Wet impingement scrubbers
  - Venturi scrubbers
  - Mechanical scrubbers
- Electrostatic precipitators

Cleaners for Gases and Vapors

- Adsorbers
- Absorbers (generally equivalent to scrubbers)
- Incinerators
- Catalytic combustion

Weather Considerations for Pollution Control

As has been stated many times, weather is something we all discuss but can do very little about. This is also true in the case of air pollution. Weather factors cannot be changed. However, it is desirable to consider weather in terms of where to locate a plant or process as well as to determine the location and size of any particular stack or vent. Weather conditions will also help to determine the vector of travel as well as the dispersion of pollutants that are emitted from the plant.

The first factor to consider when looking at weather patterns is the wind. Generally, an area has a prevailing wind: that is, a wind that blows from a particular direction the greatest percentage of the time. The speed at

which this wind travels as well as the variability of speed are important considerations in determining the dispersion of pollution. In addition to the wind, one must consider the effects of topography and the height of the pointof emission on the pollution generated. If the topography is relatively rough, then a turbulence can be expected that results in a high variability of wind direction, thus making it difficult to predict the path of the contaminant flow.

A second factor to consider when looking at the weather is the temperature lapse rate. The temperature lapse rate is defined as the rate of decrease of temperature with height. Given certain meteorological conditions, a warm blanket of air can be trapped close to the earth by a cold air blanket above; this phenomenon is called an inversion. When this situation occurs, pollutants that are introduced into the atmosphere tend to stay close to the ground and do not disperse widely, causing difficulties because of their concentration. It is during these periods of inversion that major air pollution problems have occurred.

Precipitation cleans the air, much like water sprayed in a spray-tower air cleaner. As precipitation falls, it removes particles from the air and may absorb gases or vapors in the air. These contaminants are then washed to the ground.

The actual temperature range that is experienced within an area can be of importance in air pollution problems. The range of temperature from morning to evening provides a rough index concerning the stability of the air layers. In addition, when the temperature ranges to very cold, a significant pollution effect may be expected from the operation of heating systems.

Humidity is a factor that must also be considered. If the air is too dry, the result is a loading of dust through natural causes. If the air becomes too humid, the resulting fog can hold an inversion longer by preventing the sun from heating the lower air and making it rise. This traps pollutants that can attach themselves easily to water vapors in the air.

In the same manner, cloud cover affects the temperature lapse rate and results in a higher possibility of inversion. In any case, even though there is nothing we can do about the weather, it is important that we understand those factors that can affect the rate of pollution present in a given area.

Summary

The installation of a ventilation system to remove contaminants from the workplace area does not complete the responsibility of the industrial hygiene engineer. It is necessary to assure that the contaminants are not dumped outside the plant where harmful effects to the general population and environment will result. In order to prevent such a situation from arising, an air cleaner can be attached to the ventilation system to remove contaminants before they enter the atmosphere.

In order to determine if air pollution is present, two general methods of sampling can be employed. The industrial hygiene engineer may sample the

ambient air in the community. This provides information concerning the concentration of contaminants to which various segments of the population are exposed. On the other hand, sampling of the exhaust stack can be performed which will indicate the extent of contaminants being introduced at a single source. Both methods involve essentially the same equipment but provide different information.